Experiment 9: Functional Test Case Design

> **Aim:** To design functional test cases for the Online Quiz System and link each test case to its requirement.

9.1 Requirements to test

Req. ID	Requirement in simple words	SRS reference
R-01	Encrypted user authentication and role-specific access control.	FR-01, NFR-02
R-02	Instructor question bank management and test configuration (duration, marks, rules).	FR-02, FR-03
R-03	Candidate assessment delivery (randomization, synchronized countdown, auto-save).	FR-04, FR-05, FR-06
R-04	Instant automated grading, performance analytics, and report exports.	FR-07, FR-08

9.2 Test case register

Test ID	Req.	Scenario	Expected result	Type
TC-01	R-01	Login with valid institutional credentials and correct role assignment.	Login succeeds; user is redirected to the appropriate role dashboard.	Positive
TC-02	R-01	Login with incorrect password or unregistered email ID.	Login fails; system displays an invalid credentials error message.	Negative
TC-03	R-02	Instructor adds a new multiple-choice question to the question bank.	Question is saved successfully and listed in the question repository.	Positive
TC-04	R-02	Instructor configures a quiz with invalid time window (End time before start time).	System rejects configuration and prompts for a valid time window.	Negative
TC-05	R-03	Candidate starts a quiz where questions and options are randomized.	Questions and choices render in a randomized sequence per session.	Positive
TC-06	R-03	Candidate response auto-saves asynchronously during exam navigation	Selected option state persists asynchronously without data loss	Positive

TC-07	R-03	Candidate attempts to modify answers after the countdown timer reaches 00:00:00.	System locks responses and blocks any further edits.	Negative
TC-08	R-03	Synchronized countdown timer reaches 00:00:00 during an active session.	System automatically auto-submits candidate responses and closes session.	Positive
TC-09	R-04	Candidate submits completed quiz before timer expiration.	Objective scores are calculated instantly with performance breakdown.	Positive
TC-10	R-04	Candidate attempts a second submission attempt for the same quiz without admin reset.	Submission is blocked; system enforces the single-attempt rule.	Negative
TC-11	R-04	Instructor exports cohort results and item-wise performance data.	System successfully downloads structured CSV/PDF roster report.	Positive

9.3 Result

In this experiment we wrote 11 positive and negative functional test cases for the Online Quiz System and linked each one directly to its corresponding requirement.

Overall Conclusion

In Experiments 1 to 9 we worked on our project *Online Quiz System* from initial requirements gathering to architectural design and functional testing. The SRS gave us a clear list of what the system must do—such as automated grading, randomized delivery, and strict countdown enforcement—and what constraints it must operate under. Comparing process models demonstrated why an iterative, risk-driven approach suits an educational platform where usability and data security are critical.

The use case diagram showed who interacts with the system (Administrators, Instructors, and Students), and the DFD illustrated how assessment data and session metrics move through it. The class diagram provided the main entities (User, Question, Quiz, Scorecard) and their relationships, while the sequence diagrams mapped out the exact order of messages for authentication, timed exam delivery, and asynchronous auto-saving. Modular design techniques helped us maintain high cohesion for quiz evaluation logic and low coupling between the interface and the database. Finally, the test case register linked every core requirement to an actionable testing scenario. Together, these software engineering steps make the system easier to understand, implement, and verify, proving that structural flaws are found and fixed much earlier on paper than in raw code.